EBENEZER CHRISTIAN SS NAMULANDA

BEGINNING OF TERM 1 2026

S.6 CHEMISTRY

PROPOSED GUIDE

BY TR. SSEMATIMBA GODFREY

ITEM 1: ENTHALPY CHANGES

(a) Why combustion reactions are exothermic

In a combustion reaction, chemical bonds in the reactants must first be broken before new bonds can be formed.

- Breaking bonds in the fuel and oxygen molecules **requires energy**; this energy is absorbed from the surroundings.
- When carbon dioxide and water are formed, **new bonds are created**, and bond formation **releases energy**.
- In combustion reactions, the total energy released during bond formation is **greater** than the energy absorbed during bond breaking.

Because more energy is released than absorbed, there is a **net release of energy to the surroundings**, and therefore combustion reactions are **exothermic**.

b.

$$\text{Moles of Fuel X} = \frac{2.0}{32} = 0.0625 \text{ mol}$$

$$\Delta H = \frac{-96}{0.0625} = \boxed{-1536 \text{ kJ mol}^{-1}}$$

Fuel Y

$$\text{Moles of Fuel Y} = \frac{2.0}{44} = 0.0455 \text{ mol}$$

$$\Delta H = \frac{-72}{0.0455} \approx \boxed{-1584 \text{ kJ mol}^{-1}}$$

(c) Comparison and recommendation

- **Energy per mole**:
  - Fuel Y releases slightly more energy per mole.
- **Energy per gram**:
  - Fuel X: 96/2.0=**48 kJ g−1**
  - Fuel Y: 72/2.0=**36 kJ g−1**

Fuel X releases more energy per gram of fuel and is therefore more efficient for industrial use.

Recommendation: Fuel X is more suitable for the factory.

(d) Hess's Law

Hess's Law states that the enthalpy change of a reaction depends only on the initial and final states of the system and is **independent of the pathway** taken.

This means that even if a reaction occurs in several steps, the overall enthalpy change is the same as if the reaction occurred in one step. Therefore, Hess's Law allows enthalpy changes to be calculated indirectly using known enthalpy changes of related reactions, especially when direct measurement is difficult

(e) Enthalpy change using formation enthalpies

$$\Delta H = \sum \Delta H_f(\text{products}) - \sum \Delta H_f(\text{reactants})$$

$$= [(-393) + (-286)] - (-75)$$

$$= -679 + 75 = \boxed{-604 \text{ kJ mol}^{-1}}$$

The negative value confirms that the combustion of Fuel X is exothermic.

(f) Safety and environmental hazards

• Fire **or explosion risk:** Combustion fuels are highly flammable.
Precaution: Use fire-resistant equipment and automatic fire suppression systems.

• Carbon **dioxide emissions:** Combustion releases CO_2, which contributes to global warming.
Precaution: Ensure good ventilation and use emission-control technologies

Item 2

(a) Effect of temperature

Increasing temperature increases the kinetic energy of gas molecules, causing more frequent and energetic collisions. This increases the **rate of reaction**.

However, the forward reaction is exothermic. Increasing temperature adds heat to the system. According to **Le Chatelier's principle**, the equilibrium shifts in the direction that absorbs the added heat the **reverse (endothermic) reaction**. This reduces the equilibrium yield of ammonia.

Thus, higher temperature increases reaction rate but decreases ammonia yield

(b) Methods to Increase the Yield of Ammonia

(i) Increasing pressure

The reaction goes from **4 moles of gas** on the left to **2 moles of gas** on the right.

Increasing pressure forces the system to reduce the number of gas molecules. According to Le Chatelier's principle, equilibrium shifts towards the side with fewer moles of gas the **product side**, increasing ammonia yield.

(ii) Removing ammonia as it forms

Ammonia is removed by cooling and liquefying it.

Removing ammonia lowers its concentration. The equilibrium is disturbed, and the system responds by shifting to the right to replace the removed ammonia, resulting in increased ammonia production.

(c) Safety and environmental risks

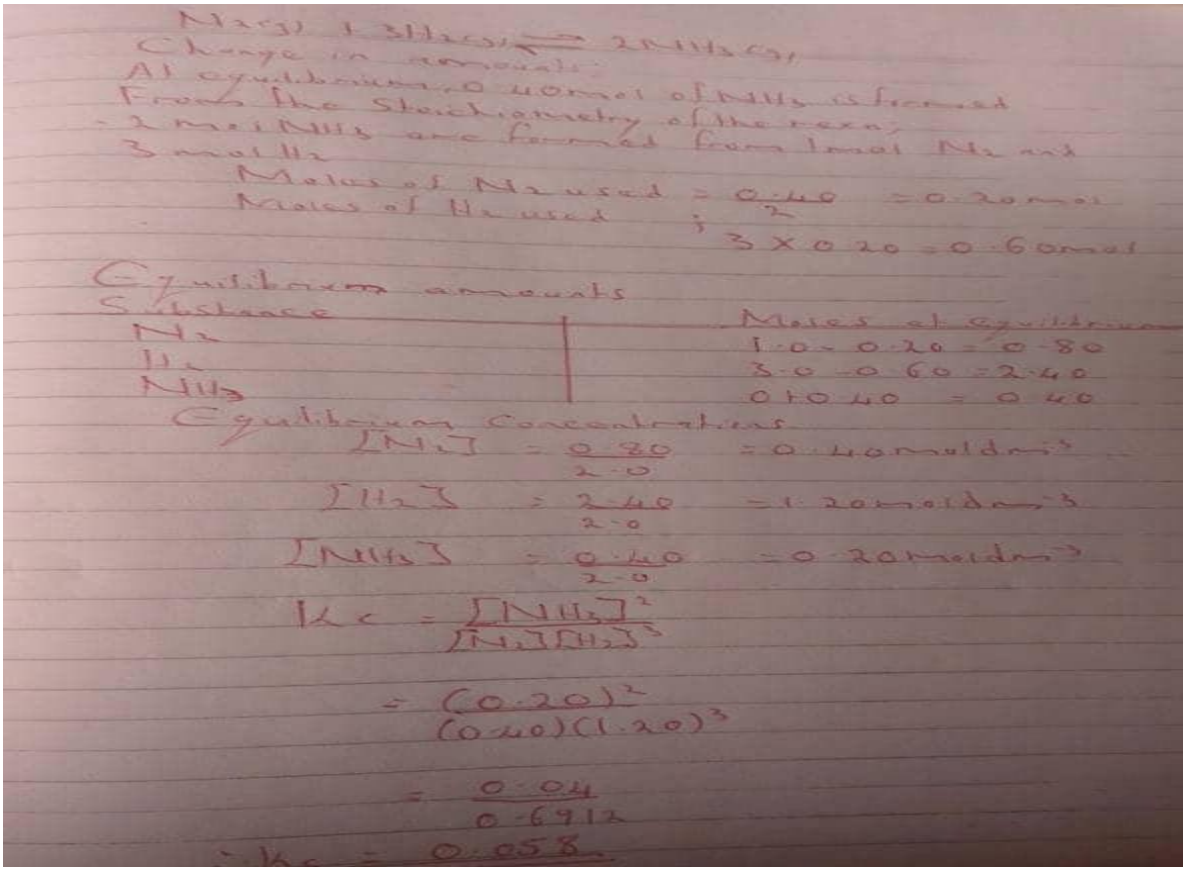
- **High pressure operation:** Risk of explosion.
  Precaution: Use reinforced vessels and pressure-relief valves.
- **Ammonia toxicity:** Ammonia is poisonous and corrosive.
  Precaution: Install gas detectors and ensure adequate ventilation.

(d) Equation and industrial conditions

$N_2 + 3H_2 \rightleftharpoons 2NH_3$

- **Conditions:** High pressure, moderate temperature, iron catalyst.

(e) Calculation of equilibrium constant, Kc



$N_2(g) + 3H_2(g) \rightleftharpoons 2NH_3(g)$

Change in amounts:

At equilibrium, 0.40 mol of NH_3 is formed

From the stoichiometry of the reaction:

2 mol NH_3 are formed from 1 mol N_2 and 3 mol H_2

Moles of N_2 used $= \frac{0.40}{2} = 0.20$ mol

Moles of H_2 used $= 3 \times 0.20 = 0.60$ mol

Equilibrium amounts

Substance	Moles at equilibrium
N_2	$1.0 - 0.20 = 0.80$
H_2	$3.0 - 0.60 = 2.40$
NH_3	$0 + 0.40 = 0.40$

Equilibrium concentrations

$[N_2] = \frac{0.80}{2.0} = 0.40$ mol dm^{-3}

$[H_2] = \frac{2.40}{2.0} = 1.20$ mol dm^{-3}

$[NH_3] = \frac{0.40}{2.0} = 0.20$ mol dm^{-3}

$$K_c = \frac{[NH_3]^2}{[N_2][H_2]^3}$$

$$= \frac{(0.20)^2}{(0.40)(1.20)^3}$$

$$= \frac{0.04}{0.6912}$$

$$\therefore K_c = 0.058$$

(f) Modified experiment

Increasing the initial concentrations of nitrogen and hydrogen shifts equilibrium towards ammonia formation. Using the same Kc, a larger amount of ammonia is present at equilibrium because more reactant molecules are available.

(g) Percentage yield

Theoretical maximum NH_3 = 0.67 mol
Actual yield = 0.40 mol

% yield = 0.40/0.67×100 ≈ **60%**

The modified experiment gives a higher yield because equilibrium is shifted towards products.

SECTION B – PART I

ITEM 3: PERIODICITY I (Atomic Structure and Trends)

The mining company analysed four elements, W, X, Y, and Z, extracted from the same period of the Periodic Table. The data provided include atomic number, atomic radius, and first ionisation energy.

(a) Analysis of trends in atomic radius and ionisation energy

Across the period from element W to element Z, the **atomic radius decreases steadily**, while the **first ionisation energy increases**.

This trend occurs because:

- As atomic number increases across a period, **the number of protons in the nucleus increases**.
- The added electrons enter the **same energy level**, so there is **no significant increase in shielding**.
- The increased nuclear charge exerts a **stronger electrostatic attraction** on the outer electrons.

As a result:

- The electron cloud is pulled closer to the nucleus, causing a **decrease in atomic radius**.
- More energy is required to remove an electron, leading to an **increase in first ionisation energy**.

(b) Identification of the most metallic element

Metallic character is associated with:

- Large atomic radius
- Low ionisation energy
- Ease of losing electrons

Element **W** has:

- The **largest atomic radius (186 pm)**
- The **lowest first ionisation energy (496 kJ mol^{-1})**

This means its outer electrons are weakly held and can be easily lost during chemical reactions. Therefore, **element W is the most metallic element** among those listed and is most suitable for applications such as electrical wiring.

(c) Element most likely to form negative ions

Non-metals tend to:

- Have small atomic radii
- Possess high ionisation energies
- Gain electrons to achieve a stable electron configuration

Element **Z** has:

- The **smallest atomic radius**
- The **highest ionisation energy**

These properties indicate a strong attraction for electrons. As a result, element Z is most likely to **gain electrons and form negative ions (anions)** in chemical reactions.

(d) Influence of nuclear charge on observed trends

Nuclear charge plays a central role in determining periodic trends.

As nuclear charge increases:

- The attraction between the nucleus and the outer electrons becomes stronger.
- Electrons are pulled closer to the nucleus, reducing atomic size.
- More energy is required to remove an electron, increasing ionisation energy.

Therefore, the observed trends in atomic radius and ionisation energy across the period are a direct result of **increasing nuclear charge with constant shielding**.

(e) Conclusion and application

Based on the data and analysis:

- Element W is best suited for electrical wiring due to its metallic nature.
- Element Z is likely to be a non-metal and to form negative ions.
- The trends observed are consistent with known periodic properties of elements across a period.

ITEM 4: PERIODICITY II (Oxides and Acid–Base Behaviour)

The laboratory investigated oxides formed by Period 3 elements and studied their reaction with water and acids/bases.

(a) Identification of Oxides A, B, and C

- **Oxide A** reacts vigorously with water to give a solution of pH 13.
  This indicates a **strongly basic oxide**, typical of **sodium oxide (Na_2O)**, which reacts with water to form sodium hydroxide.
- **Oxide B** does not react with water and has a neutral pH but reacts with both acids and bases.
  This behaviour is characteristic of an **amphoteric oxide**, such as **aluminium oxide (Al_2O_3)**.
- **Oxide C** reacts slowly with water to form an acidic solution of pH 2 and reacts with sodium hydroxide.
  This indicates an **acidic oxide**, typical of **sulfur dioxide (SO_2)** or **sulfur trioxide (SO_3)**.

(b) Change in acid–base character across Period 3

Across Period 3:

- Oxides change from **basic → amphoteric → acidic**.

This trend occurs because:

- Metallic character decreases across the period.
- Elements on the left form **ionic oxides**, which contain oxide ions (O^{2-}) that react with water to form hydroxide ions.
- Elements on the right form **covalent oxides**, which react with water to produce acids.

(c) Relationship between bonding and chemical behaviour

- **Ionic oxides** (e.g. Na_2O) dissolve in water to produce alkaline solutions because oxide ions react with water.
- **Amphoteric oxides** (e.g. Al_2O_3) have both ionic and covalent character, allowing them to react with both acids and bases.
- **Covalent oxides** (e.g. SO_2, SO_3) dissolve in water to form acids.

Thus, the type of bonding directly determines the chemical behaviour of the oxide.

(d) Environmental impact of Oxide C

Oxide C forms an acidic solution when dissolved in water. If released into the environment:

- It would lower soil pH.
- Acidic soil reduces nutrient availability to plants.
- Crop growth and soil microorganisms would be adversely affected.

This explains why acidic oxides contribute to environmental problems such as **acid rain**.

(e) Conclusion

The observed behaviour of the oxides reflects periodic trends in metallic character, bonding, and acid–base properties across Period 3

ITEM 5: ORGANIC I

Two hydrocarbons, M and N, both contain four carbon atoms but show different chemical behaviours.

(a) Identification of homologous series

- Compound **M** has molecular formula C_4H_{10} and does not decolourise bromine water. This identifies it as an **alkane**, which is a saturated hydrocarbon.
- Compound **N** has molecular formula C_4H_8 and decolourises bromine water. This indicates the presence of a **carbon–carbon double bond**, identifying it as an **alkene**.

(b) Explanation of bromine water test

Bromine water is used to test for unsaturation.

- Alkanes do not react with bromine water under normal conditions, so there is no colour change.
- Alkenes undergo **addition reactions** with bromine across the C=C bond, causing the brown colour to disappear.

This explains why compound N decolourises bromine water while compound M does not.

(c) Explanation of combustion behaviour

- Compound M burns with a **clean blue flame**, indicating complete combustion and a higher hydrogen-to-carbon ratio.
- Compound N burns with a **smoky yellow flame** due to incomplete combustion and a higher carbon content, producing soot.

(d) Suitability for industrial chemical synthesis

Alkenes are more suitable for synthesis because:

- The C=C bond is reactive.
- Alkenes can undergo addition and polymerisation reactions.

Therefore, **compound N** is more suitable for industrial chemical synthesis.

(e) Industrial uses

- **Alkanes**: fuels such as LPG and petrol.
- **Alkenes**: manufacture of plastics, alcohols, and detergents.

ITEM 6: ORGANIC II

(a) Identification of functional groups

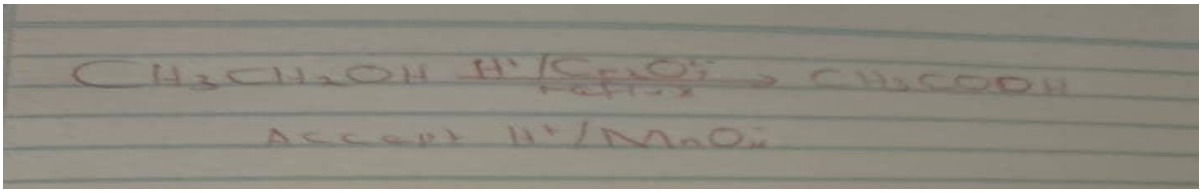
- Liquid **P** reacts with sodium metal and turns potassium dichromate green, indicating the presence of an **alcohol (–OH) group**.
- Liquid **Q** does not react with sodium metal, does not reduce dichromate, and is acidic, indicating a **carboxylic acid (–COOH) group**.

(b) Explanation of test reactions

- Sodium reacts with alcohols to produce hydrogen gas, causing effervescence.
- Alcohols are oxidised by acidified potassium dichromate, turning it from orange to green.
- Carboxylic acids do not undergo further oxidation under these conditions.

(c) Conversion of Liquid P to Liquid Q

Liquid P (ethanol) can be oxidised to Liquid Q (ethanoic acid):



This involves:

- Oxidation of the alcohol group
- Increase in oxidation state of carbon

(d) Industrial or medicinal use of Liquid Q

Carboxylic acids such as ethanoic acid are used:

- As food preservatives
- In pharmaceutical synthesis
- In the manufacture of esters

(e) Conclusion

The test results clearly distinguish between alcohols and carboxylic acids, demonstrating how functional groups determine chemical behaviour and practical applications

END